Microsoft Elevate

CAPSTONE PROJECT

Bike Rental Demand Analysis using Power BI

PRESENTED BY

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OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result
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PROBLEM STATEMENT:

Urban transportation systems are increasingly adopting bike rental services as an eco-friendly and convenient mobility solution. However, managing bike availability according to demand is a significant challenge for service providers. Demand varies depending on factors such as **time of day, season, and environmental conditions**. Without proper analysis of historical data, it becomes difficult to identify peak demand periods and optimize resource allocation. Therefore, analyzing bike rental data is essential to understand usage patterns and support better operational planning.

PROPOSED SOLUTION:

The project analyzes bike rental data using **data visualization techniques** to understand demand patterns.

Proposed Approach:

- Collect historical bike rental dataset
- Clean and prepare data for analysis
- Create interactive dashboards using **PowerBI**
- Identify peak rental hours and seasonal trends
- Provide insights through data visualization

SYSTEM APPROACH:

The system is developed using **Microsoft Power BI**, a powerful business intelligence tool used for data analysis and visualization.

❖ **Technologies Used**

- Microsoft Power BI
- Data Cleaning and Transformation
- Data Visualization
- Interactive Dashboard Development

These tools help convert raw data into meaningful insights through charts, graphs, and KPIs.

ALGORITHM & DEPLOYMENT:

The following steps were followed during the development of this project:

- Collection of historical bike rental dataset from a reliable data source
- Import the dataset into **Microsoft Power BI** for analysis
- Perform **data cleaning and preprocessing** to remove missing or inconsistent values
- Transform the dataset using **Power Query** for better structure and usability
- Create **calculated columns and DAX measures** to derive useful metrics
- Design various **data visualizations such as charts, graphs, and KPIs**
- Build an **interactive dashboard** to display bike rental trends and patterns
- Analyze rental demand based on **hour, season, and environmental factors**
- Deploy the final dashboard for **data interpretation and decision-making**

RESULT:

The developed dashboard provides insights into bike rental demand through various visualizations such as hourly rental trends, seasonal patterns, and overall rental distribution. These insights help understand usage behavior and peak demand periods.

Hourly Bike Rental Demand Analysis

Peak Demand Hour

23

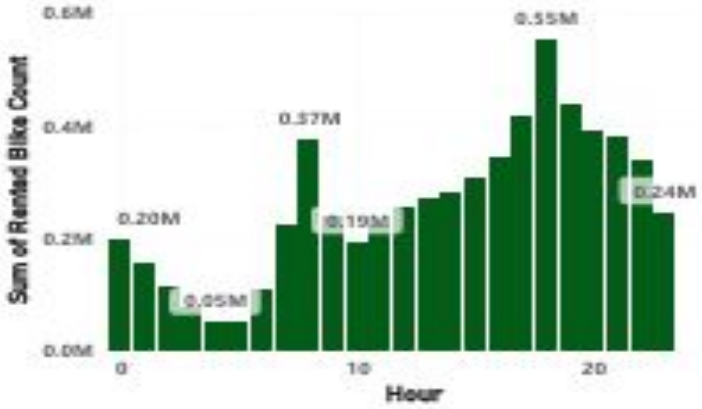
Peak Hour Rentals

3556

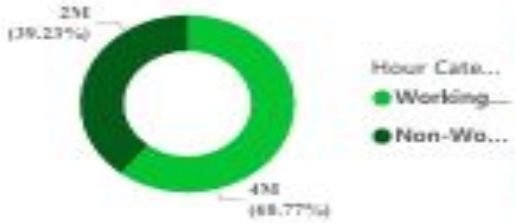
Avg Hourly Rentals

704.60

Hourly Bike Demand Pattern



Rentals During Working vs Non-Working Hours



Bike Demand Heatmap by Hour & Day

Hour		1	2	3	4	5	6	7
0	123213	1764	1590	2388	4219	7536	11531	887
1	98258	1665	1503	1948	3053	5125	8824	730
2	69793	1289	1086	1468	1944	3694	6070	527
3	46641	781	739	987	1324	2568	4161	368
4	30161	522	478	626	880	1610	2899	236
5	31284	650	438	680	949	1658	3169	256
6	64027	1234	974	1350	2421	3472	6672	507
7	137038	2696	2167	3537	6465	7705	13273	940
8	231178	5537	4425	6101	11599	12749	20231	1542
Total	3843810	64062	53694	117813	170431	235021	371295	27295

Demand Distribution Across the Day

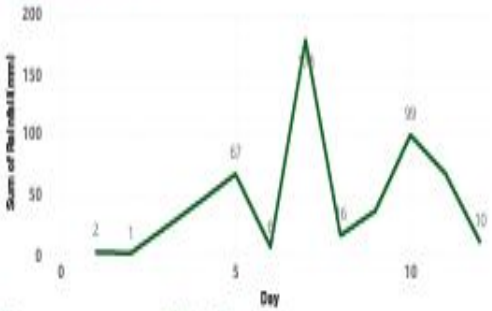
KEY INSIGHTS

- *Peak bike demand occurs between 6 PM – 8 PM*
- *Morning commuting hours show steady demand*
- *Late night hours have minimal rental activity*
- *Weekday commuting hours dominate rental usage*

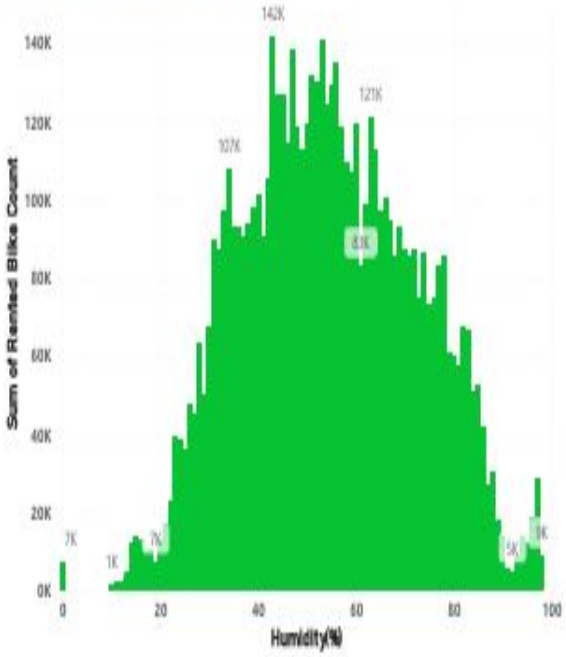
Weather Impact on Bike Rental Demand

Avg Temperature	Avg Humidity	Total Rainfall
4.07	58.23	1.30K

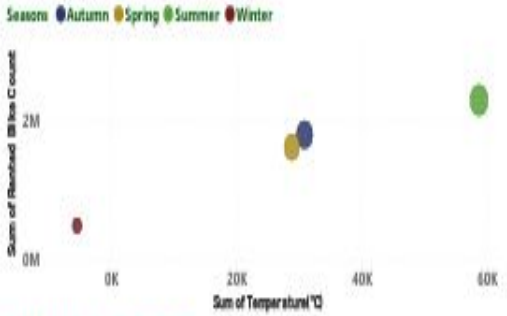
Rainfall Trend



Bike Rentals vs Humidity



Temperature vs Bike Rentals



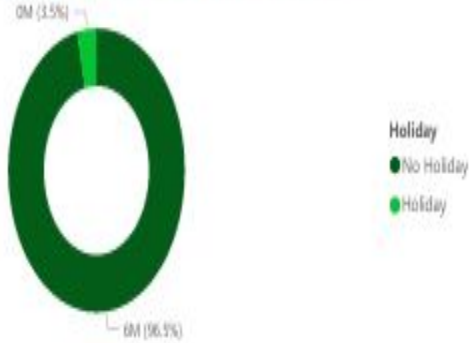
WEATHER INSIGHTS

- *Moderate temperatures lead to higher bike rentals*
- *High humidity slightly reduces bike usage*
- *Rainfall significantly decreases rental activity*
- *Weather plays a major role in bike demand patterns*

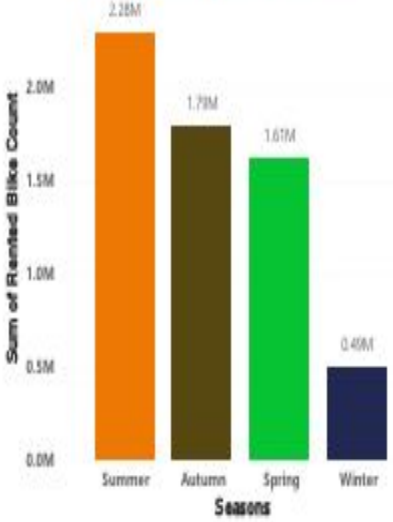
Seasonal and Holiday Bike Rental Trends

216K	6M
Holiday Rentals	Non-Holiday Rentals

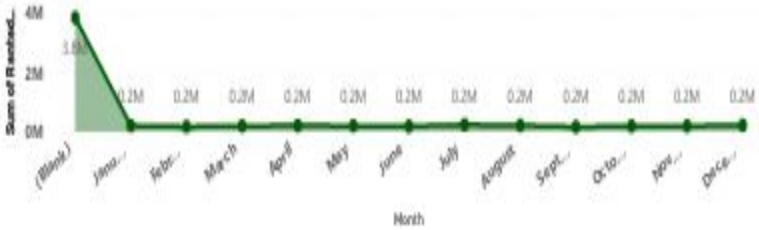
Bike Rentals: Holiday vs Non-Holiday



Season-wise Bike Rentals



Monthly Bike Rental Trend



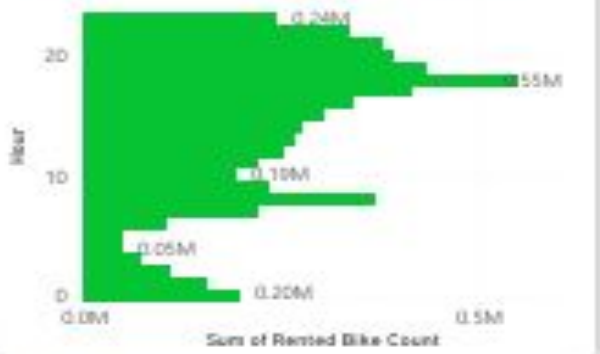
SEASONAL INSIGHTS

- *Summer season records the highest bike rentals*
- *Winter shows the lowest rental activity*
- *Non-holiday working days dominate bike usage*
- *Bike demand increases during warmer months*

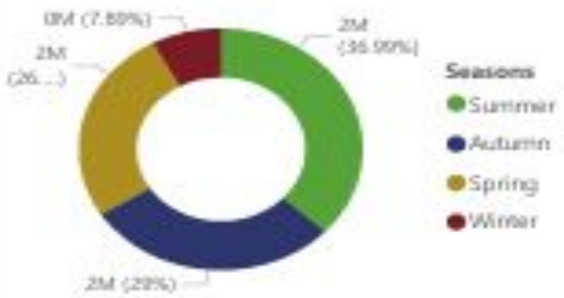
Bike Rental Demand – Key Insights

6M	3556	705	4.07
Total Rentals	Peak Rentals	Average Rentals	Avg Temperature

Top Demand Hours Chart



Season Demand Distribution



Business Recommendations

- *Increase bike availability during peak hours (6 PM – 8 PM)*
- *Deploy more bikes during summer season*
- *Reduce maintenance downtime during high demand months*
- *Improve bike availability during weekday commuting hours*

Seasons

Autumn	Summer
Spring	Winter

KEY INSIGHTS

- *Peak bike demand occurs between 6 PM – 8 PM*
- *Summer season records the highest rental activity*
- *Rainfall significantly reduces bike usage*
- *Weekday commuting hours dominate rental demand*

CONCLUSION:

The analysis of bike rental data provides valuable insights into customer usage patterns and demand fluctuations. By analyzing historical data using data visualization techniques, important trends related to time, season, and environmental conditions can be identified. The interactive dashboard developed using **Microsoft Power BI** enables users to clearly understand rental demand patterns through visual charts and indicators.

Through this project, raw data has been transformed into meaningful insights that help in identifying peak rental hours, seasonal demand variations, and overall usage behavior. These insights can assist service providers in improving resource allocation, ensuring better availability of bikes, and enhancing operational efficiency. Overall, the use of business intelligence tools plays a significant role in supporting data-driven decision making and improving the effectiveness of urban mobility services.

FUTURE SCOPE:

Future enhancements for this project may include:

- ★ Integration with **real-time bike rental data** to monitor demand dynamically
- ★ Implementation of **predictive analytics models** to forecast future rental demand
- ★ Use of **machine learning techniques** to improve demand prediction accuracy
- ★ Development of **location-based analysis** to identify high-demand areas
- ★ Enhancement of the dashboard with **mobile-friendly access and automated reporting**

REFERENCES:

- Bike Sharing Dataset downloaded from **Kaggle** for data analysis and visualization.
- Dashboard development and data analysis performed using **Microsoft Power BI**.
- Learning resources and study materials referred from **Microsoft Learn**.
- Additional insights and practical understanding of Power BI gained through technical live sessions conducted during the internship program.
- Official documentation and tutorials from *Microsoft Power BI* platform.

GitHub Link:
https://github.com/s-s-shriram/Bike-Rental_Demand_Analysis_using-Microsoft_Power-BI.git

Thank You